

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. Branch: Information Technology Semester: III

Subject Code & Name: BTITC305 Data Structures and Applications

Max Marks: 60

Date:11/01/2024

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) What is the complexity of an algorithm? Define asymptotic notations for complexity of algorithms.	Understand Knowledge CO3	6
B) Describe the notion of the space-time tradeoff of algorithms. Calculate time complexity of the following function. int data(int list[], int n) { int i, sum= 0; for(i=0; i<n; i++) sum = sum + list[i] ; return sum ; }	Knowledge Apply CO3	6
C) Write an algorithm: i)To add two n x n matrices ii)To find the largest element in the list	Apply CO1	6
Q.2 Solve Any Two of the following.		12
A) Evaluate the following postfix expression using stack. 5, 6, 2, +, *, 12, 4, /, -	Evaluate CO2	6
B) What is a priority queue? Describe an array representation of a priority queue.	Knowledge Understand CO1	6
C) Design a procedure which deletes a node N with location LOC in a linked list.	Create CO1	6

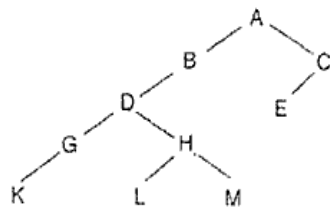
Q.3 Solve Any Two of the following.

12

- A) Draw the tree structure for the following expression.
 $(x-4y)(6a-4b)^3$
- B) Write a stack based algorithm for postorder traversal of a binary tree.
 Apply it on the following binary tree, show the steps.

Apply
 CO1
 Create
 Apply
 CO3

6



6

- C) Define binary search tree.
 Insert the following numbers in order into an empty binary search tree,
 show the stages.
 40, 60, 50, 33, 55, 11

Knowledge
 Create
 CO1

6

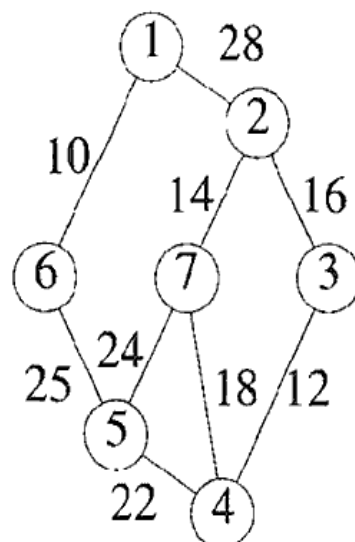
Q.4 Solve Any Two of the following.

12

- A) Describe the following terms with reference to Graph.
 i) Multigraph
 ii) Strongly Connected Graph
 iii) Path Matrix
- B) Find the minimum spanning tree using Prim's algorithm for the following graph.

Understand
 CO1

6



Apply
 CO3

6

- C) Explain linked representation of a graph with a suitable example.

Understand
 CO3

6

Q. 5 Solve Any Two of the following.

12

- | | | |
|---|---------------|---|
| A) Write a C program to implement a Binary Search algorithm for searching a given item in the list. | Create
CO4 | 6 |
| B) Sort the following list using Selection Sort, show the passes.
77, 33, 44, 11, 88, 22, 66, 55 | Apply
CO3 | 6 |
| C) Write a C program to implement a Quick Sort algorithm for sorting a given list. | Create
CO5 | 6 |

***** End *****